Acoustic Emission and Passive Acoustic Regression to the ISO 281 Viscosity Ratio κ for Online, Non-Invasive Lubrication Condition Monitoring of Ball Bearings: An Exploratory Study

Frederik Appel Vardinghus-Nielsen

Department of . . . , Aarhus University, Denmark
frederikavn@gmail.com

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Abstract

Inadequate lubrication is responsible for an estimated 30–80% of rolling element bearing failures, yet reliable online, non-invasive lubrication condition monitoring (LCM) remains an unsolved engineering problem. Passive sensor-based approaches are most practical for industrial retrofit because they require no bearing modification, but the signal processing challenge of separating lubrication-relevant information from confounded operating-condition effects has not been systematically characterised for simultaneous acoustic emission (AE) and passive ultrasound (UL) sensing. This paper presents an exploratory study in which AE and passive UL signals are simultaneously acquired from a ball bearing during a continuous speed sweep (1–5985 rpm) at temperatures spanning 33–100 °C, covering the boundary, mixed, and full-film lubrication regimes across 8 783 sweep–sensor pairs per channel. The ISO 281 viscosity ratio $\kappa \in [0.003, 1.64]$ (mean 0.51) is derived analytically from bearing geometry, lubricant properties, and operating conditions as a continuous, physics-grounded regression target. For the AE channel, 26 time-domain and frequency-domain features are extracted from the broadband signal and from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), yielding 104 candidate features; 26 features are extracted from the pre-filtered UL channel (0–20 kHz). A two-stage selection retains 32 AE and 6 UL features after a joint Spearman–Pearson correlation threshold filter and a greedy variance-inflation-factor and pairwise-correlation redundancy reduction. Three regression model families are benchmarked across three sensor configurations using repeated nested five-fold cross-validation ($R = 5$ repetitions, 25 outer scores per model); LightGBM on AE features alone achieves the best hold-out performance ($R^2 = 0.951$, RMSE $= 0.078$ κ-units). Fusing AE and UL features yields only a marginal improvement (ΔRMSE $= 0.0011$) that is not confirmed by holdout significance testing (Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap 95% CI includes zero). Significance of pairwise model differences is assessed by Nadeau–Bengio corrected repeated k-fold t-tests with Holm–Bonferroni correction for within-sensor comparisons, and by Wilcoxon signed-rank, Diebold–Mariano, and bootstrap ΔRMSE confidence intervals for cross-sensor comparisons. The dataset is made publicly available to support standardised comparison.

Contents

1 Introduction

Rolling element bearings are among the most prevalent mechanical components in industrial machinery, and their reliable operation depends critically on adequate lubrication. Lubrication failure—whether caused by starvation, viscosity degradation, or contamination—increases metal-to-metal asperity contact, accelerates surface fatigue and wear, raises operating temperature, and ultimately leads to premature bearing failure. Improper lubrication is estimated to account for between 30% and 80% of all rolling element bearing failures [1–3].

Despite this, lubrication condition monitoring (LCM) has received substantially less research attention than vibration-based fault detection and remaining useful life estimation, which are now mature technologies deployed widely in industry. The principal reason is that lubrication-related signal changes are subtler and more entangled with operating conditions than the impulse signatures of surface spalling or raceway cracks. Accurate, online LCM would enable condition-based relubrication: applying lubricant at the right time in the right quantity rather than following time-based schedules that systematically produce both under- and over-lubrication [4].

Find reference for this

Passive, non-invasive sensing methods – acoustic emission (AE), vibration, and passive ultrasound (US) – are practically attractive for industrial LCM because they require no bearing modification and can be retrofitted to bearings in service. Among these, AE is sensitive to the broadband stress waves emitted by asperity contact events and micro-crack propagation across a wide frequency range (typically 50 kHz–4 MHz), while passive ultrasound at 38–40 kHz is already industrially deployed for threshold-based lubrication assessment [5]. Despite both being in practical use, they have not been studied simultaneously in a unified regression framework with a physics-based lubrication reference variable.

Find reference for this

The ISO 281 viscosity ratio κ has been used as a continuous regression target [1, 6] to develop models, which are sensitive to lubrication changes. Since κ encodes the ratio of actual lubricant viscosity to the minimum required for full elastohydrodynamic lubrication (EHL) at the current operating conditions, a model trained to predict κ implicitly learns the operating-condition dependence and provides a physically interpretable, standardised output. Jakobsen et al. [1, 6] demonstrate κ regression from vibration and passive ultrasound features using LASSO and neural networks, establishing the methodological foundation that this paper builds upon.

The complementarity of AE and passive US for LCM has been argued on physical grounds in several reviews [7, 8], but has not been quantified experimentally in a unified framework. Whether jointly exploiting both sensor modalities improves κ prediction over either alone is an open question with direct practical implications: additional sensors impose cost and complexity, and are only justified if they provide non-redundant information.

Contributions

This paper makes the following contributions:

1. **Novel dual-channel experiment.** The first study to simultaneously acquire AE and passive US signals from a ball bearing and regress both, individually and jointly, to κ using a common dataset and analytical ground truth.

2. **Sub-band feature extraction for AE.** Features are computed from the broadband AE signal and from three frequency sub-bands (20–500 kHz, 500 kHz–1 MHz, 1–2 MHz), enabling model-driven identification of the most informative AE frequency range for lubrication state prediction.

3. **Quantified fusion test.** The first quantitative test of AE–US complementarity in a unified experimental framework, including a positive fusion result that directly addresses an open question in the LCM literature.

4. **Rigorous statistical evaluation.** Model comparisons are grounded in repeated nested cross-validation and corrected significance tests (Nadeau–Bengio, Holm–Bonferroni, Wilcoxon, Diebold–Mariano, bootstrap ΔRMSE) that properly account for data re-use in cross-validated estimates.

The remainder of the paper is organised as follows. Section 2 reviews the theoretical background of κ and the physical origins of AE and UL signals. Section 3 describes the experimental setup and data collection. Section 4 presents the signal processing and feature extraction pipeline. Section 5 presents the feature analysis results. Section 6 presents the regression modelling and statistical evaluation. Section 7 discusses the findings. Section 8 concludes.

2 Background

2.1 Lubrication regimes and the viscosity ratio κ

The lubrication regime of a rolling element bearing is determined by the ratio of the elastohydrodynamic (EHL) film thickness h_m to the composite surface roughness σ of the contact surfaces, quantified by the specific film thickness $\Lambda = h_m/\sigma$. When $\Lambda \geq 3$ the contact is in the full-film EHL regime; when $0.5 \leq \Lambda < 3$ mixed lubrication prevails with partial asperity contact; when $\Lambda < 0.5$ boundary lubrication dominates with predominantly metal-to-metal contact.

The ISO 281 standard defines the viscosity ratio

$$\kappa = \frac{\nu_0}{\nu_1(n,\, d_{pw})}, \tag{1}$$

where ν_0 [$\mathrm{mm^2\,s^{-1}}$] is the actual kinematic viscosity of the lubricant at operating temperature, and ν_1 is the minimum kinematic viscosity required for full EHL at the bearing's rotational speed n [rpm] and mean pitch diameter d_{pw} [mm] [9]. The reference viscosity is given by

$$\nu_1 = \begin{cases} 45\,000\; n^{-0.83}\; d_{pw}^{-0.5} & n < 1000\,\mathrm{rpm}, \\ 4\,500\; n^{-0.5}\; d_{pw}^{-0.5} & n \geq 1000\,\mathrm{rpm}. \end{cases} \tag{2}$$

The operating viscosity ν_0 is computed from the ASTM D341 Walther viscosity–temperature equation fitted to the lubricant's 40 °C and 100 °C reference values. A value of $\kappa \geq 1$ indicates adequate film formation; $\kappa < 0.5$ corresponds to severe boundary lubrication.

Since κ encodes both n and T through ν_1 and ν_0, it cannot be varied independently of operating conditions without changing the lubricant itself. Using κ as a regression target—rather than a discrete fault label—means that a trained model implicitly compensates for operating-condition effects: the model learns the full mapping from raw features to the physics-grounded EHL adequacy index.

2.2 Physical origins of AE and passive ultrasound signals

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Acoustic emission (AE) refers to transient elastic stress waves generated by sudden energy releases in a material: asperity contact and micro-fracture are the dominant sources in rolling bearings. Under boundary and mixed lubrication, intermittent metal-to-metal asperity contacts generate broadband stress waves that propagate through the bearing structure and are detected by piezoelectric AE sensors coupled to the outer housing. The emission rate, amplitude, and frequency content of these events are all sensitive to the contact severity and hence to the lubrication regime [10, 11]. Full-film EHL suppresses asperity contact, reducing AE activity; the transition through the mixed regime is associated with intermittent, structurally distinct emission events. AE sensors typically operate in the 50 kHz–1 MHz range and require sampling rates exceeding 500 kHz to capture the full frequency content.

Passive ultrasound sensing at 38–40 kHz provides a complementary, lower-frequency view of bearing acoustics. Commercial instruments (UE Systems, Sonotec) compare the broadband ultrasonic level against a reference baseline, triggering relubrication when the level rises by a specified number of decibels above the reference [5]. This threshold approach is the only passive non-invasive LCM method in routine industrial deployment but is confounded by speed and load changes. In the present experiment the passive UL channel uses a heterodyned probe, which frequency-shifts the signal into the audible range and effectively limits the usable bandwidth to approximately 10 kHz; this constitutes a different signal from wideband AE and from classical ultrasonic reflectometry.

The two sensor modalities are therefore sensitive to partially overlapping and partially distinct physical phenomena. AE captures high-frequency transient stress waves from individual asperity contacts, while the passive UL channel reflects lower-frequency bulk acoustic behaviour and machinery noise components. Whether this partial independence translates into complementary κ information is the central empirical question of this study.

3 Experimental Setup

3.1 Test rig and bearing

The test rig is a custom-built bearing test stand at CeramicSpeed (Holstebro, Denmark). A single deep-groove ball bearing (model SYJ25, bore 25 mm, pitch diameter d_{pw} = 38.0 mm, 9 chrome steel rolling elements, dynamic load rating 14000 N) is driven by a variable-frequency electric motor controlled through a variable-frequency drive (VFD). The bearing is loaded radially by a static weight and lubricated by a defined quantity of Keratech 22 oil (kinematic viscosity $\nu_{40} = 22$ cSt, $\nu_{100} = 4.1$ cSt). Temperature is monitored by a PT100 sensor on the bearing housing and controlled passively through the test duration and ambient conditions.

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Do we need NLGI cosistency grade?

Note on VFD noise: Harmonic artefacts from the VFD switching frequency are present at approximately 5 kHz and its multiples in the low-frequency range of both sensor signals. A replacement VFD with filter is planned for subsequent experiments; in the present analysis, low-frequency spectral features below 10 kHz should be interpreted with this artefact in mind.

3.2 Sensor configuration and data acquisition

Two sensors are acquired simultaneously:

AE sensor. A wideband piezoelectric AE sensor is bonded to the bearing housing. Sampling rate: 1.6 MHz (nominally; effective bandwidth verified to 800 kHz by the DAQ). Signal recorded to a digital oscilloscope with voltage range adapted to the signal level (500 mV–1 V per division depending on RPM; see Table **??**).

Passive UL sensor. A heterodyned passive ultrasound probe (commercial instrument) is clamped to the bearing housing. The heterodyning process frequency-shifts the raw acoustic signal into the audible range (<10 kHz); the recorded signal therefore reflects low-frequency acoustic emissions rather than true wideband ultrasound. Voltage range: 50 mV per division. Sampling rate: 1.6 MHz (recorded at the same rate as the AE channel; effective content below 10 kHz).

Data are acquired in individual time-series sweeps triggered at each operating condition. Each sweep comprises a fixed number of samples from both channels recorded simultaneously. The two sensors share identical acquisition timing, ensuring that feature comparisons and correlation analyses are conducted on co-temporal signal segments.

3.3 Operating conditions and κ derivation

The experiment follows a continuous speed-sweep protocol: the VFD ramps the shaft speed from rest up to approximately 6000 rpm and back across a single extended session, with bearing temperature rising naturally from ambient (33 °C) to 100 °C as a function of speed and run time. Sweeps are acquired continuously at approximately 0.6-second intervals throughout the ramp, yielding a dense, quasi-continuous coverage of the RPM–temperature operating plane rather than a grid of discrete set-points. After signal quality filtering, 8 783 sweeps per sensor channel are retained from a single HDF5 session file.

Table 1 summarises the operating envelope and resulting κ distribution.

Table 1: Operating envelope and κ distribution for the retained dataset (8 783 sweeps per sensor). Regime boundaries follow ISO 281: $\kappa < 0.5$ (boundary), $0.5 \leq \kappa < 1.0$ (mixed), $\kappa \geq 1.0$ (full film).

Quantity	**Min**	**Max**
Shaft speed [rpm]	1	5 985
Bearing temperature [°C]	33	100
Viscosity ratio κ	0.003	1.644
Regime	**Sweeps**	**%**
Boundary ($\kappa < 0.5$)	5 281	60%
Mixed ($0.5 \leq \kappa < 1.0$)	2 552	29%
Full film ($\kappa \geq 1.0$)	950	11%
$\bar{\kappa}$	0.508	
σ_κ	0.353	

The κ value for each sweep is computed according to Equations (1)–(2) using the per-sweep measured RPM and temperature, with ν_0 evaluated via the ASTM D341 Walther viscosity–temperature equation. No separate reference measurement of film thickness is available; κ is therefore a model-derived quantity subject to the assumptions of the Hamrock–Dowson framework and Walther's equation.

Note on regime coverage. The boundary lubrication regime ($\kappa < 0.5$) dominates the dataset (60%) because the bearing spends the largest fraction of the speed ramp at lower speeds where the required viscosity ν_1 is high. Full-film conditions ($\kappa \geq 1.0$) are reached only at higher shaft speeds and account for 11% of sweeps. Model performance in the full-film regime is therefore evaluated on a smaller absolute number of sweeps, which is a limitation discussed in Section 7.

3.4 Dataset availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the analysis pipeline scripts) will be made publicly available on Zenodo upon acceptance. The Python analysis pipeline is configured via a single YAML file and is fully reproducible from the raw HDF5 files.

4 Signal Processing Pipeline

In this section the signal processing pipeline applied to each sweep–sensor pair, comprising signal cleaning, and feature extraction.

- Goal of the pipeline
- Elements included in the pipeline
- Practical implementation (Python version, libraries used, etc.)

4.1 Signal cleaning

4.2 Filter bands

4.3 Feature extraction

Twenty-six features are computed from the cleaned voltage signal for each sweep–sensor pair, comprising 12 time-domain and 14 frequency-domain statistics, as listed in Tables 2 and 3.

All frequency-domain features are computed from the one-sided FFT magnitude spectrum of the cleaned signal.

AE sub-band decomposition. In addition to broadband features, three frequency sub-bands are carved from the AE channel using zero-phase Butterworth bandpass filters: (i) 20–500 kHz, capturing the transition from audible to ultrasonic contact dynamics; (ii) 500 kHz–1 MHz, the primary stress-wave propagation band; and (iii) 1–2 MHz, covering high-frequency micro-fracture events. The full 26-feature set is extracted independently for each sub-band, yielding $26 \times 4 = 104$ candidate AE features (broadband plus three sub-bands).

UL channel pre-filtering. The UL channel is pre-filtered to 0–20 kHz by a zero-phase low-pass Butterworth filter before broadband feature extraction, consistent with the effective bandwidth of the heterodyned probe. No frequency sub-bands are computed for the UL channel, giving 26 candidate UL features.

Table 2: Time-domain features. Notation: N signal length; $\bar{x}$ sample mean; σ population standard deviation.

Name	**Formula**	**Description**
Peak	$(\max x - \min x)/2$	Half the peak-to-peak range.
RMS	$\sqrt{N^{-1}\sum_n x_n^2}$	Root mean square amplitude.
Std	$\sqrt{N^{-1}\sum_n (x_n - \bar{x})^2}$	Population standard deviation.
Variance	$N^{-1}\sum_n (x_n - \bar{x})^2$	Population variance (square of Std).
Skewness	$N^{-1}\sum_n [(x_n - \bar{x})/\sigma]^3$	Third standardised central moment; amplitude asymmetry.
Kurtosis	$N^{-1}\sum_n [(x_n - \bar{x})/\sigma]^4$	Fourth standardised central moment; tail heaviness.
Crest factor	$x_{\text{peak}}/x_{\text{RMS}}$	Peak-to-RMS ratio; sensitive to impulsive events.
Shape factor	$x_{\text{RMS}}\,/\,(N^{-1}\sum_n \lvert x_n\rvert)$	Ratio of RMS to mean absolute value.
Impulse factor	$x_{\text{peak}}\,/\,(N^{-1}\sum_n \lvert x_n\rvert)$	Ratio of peak to mean absolute value.
Margin factor	$x_{\text{peak}}\,/\,(N^{-1}\sum_n \sqrt{\lvert x_n\rvert})^2$	Peak divided by squared mean square-root amplitude.
Hjorth mobility	$[\mathrm{Var}(x')/\,\mathrm{Var}(x)]^{1/2}$	Square root of first-derivative to signal variance ratio; proportional to mean frequency.
Hjorth complexity	$M(x')/M(x)$	Ratio of first-derivative mobility M to signal mobility; bandwidth proxy.

5 Feature Analysis

5.1 Correlation with κ

Table 4 lists the top ten features for each sensor ranked by $|\rho(\text{feature}, \kappa)|$ across all 799 sweeps per sensor.

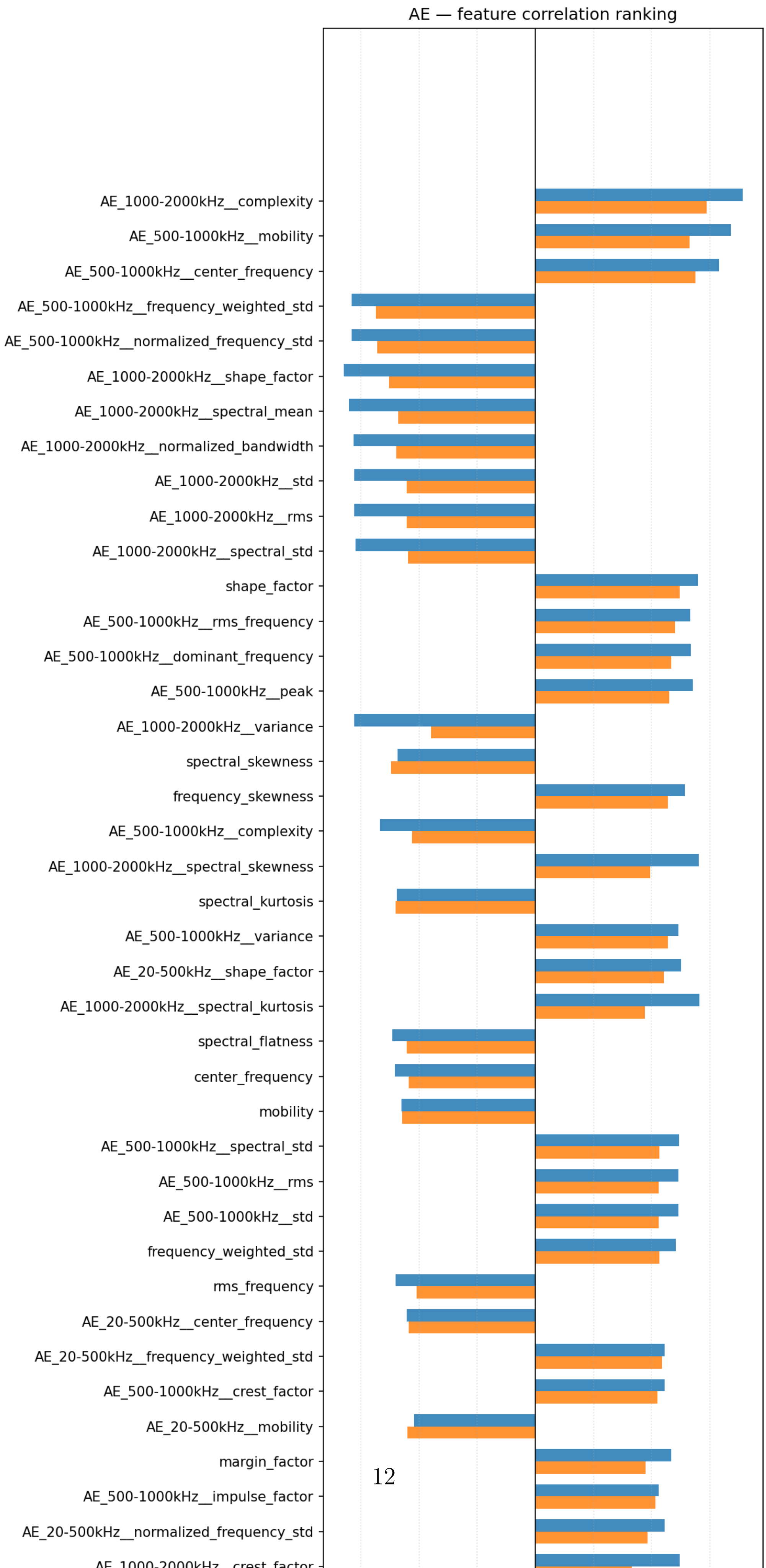
Table 3: Frequency-domain features (companion to Table 2). Notation: $S_k = |X_k|$ one-sided FFT magnitude at bin k; f_k corresponding frequency; K bin count; $\bar{S} = K^{-1} \sum_k S_k$; $\sigma_S = \mathrm{std}(\{S_k\})$; $\Sigma S = \sum_k S_k$; $f_c = (\Sigma S)^{-1} \sum_k f_k S_k$; $\sigma_f = (K^{-1} \sum_k (f_k - f_c)^2)^{1/2}$.

Name	**Formula**	**Description**
Dominant frequency	$f_{\arg\max_k S_k^2}$	Frequency at which spectral power is maximum.
Spectral mean	$\bar{S} = K^{-1} \sum_k S_k$	Arithmetic mean of the FFT magnitude spectrum.
Spectral std	$\sigma_S = \mathrm{std}(\{S_k\})$	Population standard deviation of the FFT magnitude spectrum.
Spectral skewness	$\sum_k (f_k - f_c)^3 S_k \,/\, (\sigma_S^3 \, \Sigma S)$	Frequency asymmetry of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral kurtosis	$\sum_k (f_k - f_c)^4 S_k \,/\, (\sigma_S^4 \, \Sigma S)$	Frequency tail heaviness of the magnitude-weighted spectrum, normalised by magnitude std.
Spectral flatness	$\exp(K'^{-1} \sum_{k:S_k>0} \ln S_k) \,/\, \bar{S}$	Geometric-to-arithmetic mean ratio; 1 for white noise, 0 for a pure tone.
Centre frequency	$f_c = \sum_k f_k S_k \,/\, \Sigma S$	Magnitude-weighted mean frequency (spectral centroid).
RMS frequency	$(\sum_k f_k^2 S_k \,/\, \Sigma S)^{1/2}$	Magnitude-weighted root mean square frequency.
Freq.-weighted std	$\sigma_f = (K^{-1} \sum_k (f_k - f_c)^2)^{1/2}$	RMS deviation of bin frequencies from centre frequency (unweighted by magnitude).
Peak frequency	$(\sum_k f_k^4 S_k \,/\, \sum_k f_k^2 S_k)^{1/2}$	RMS frequency computed with $f_k^2 S_k$ weighting.
Norm. freq. std	$\sigma_f \,/\, f_c$	Frequency standard deviation normalised by centre frequency.
Frequency skewness	$\sum_k (f_k - f_c)^3 S_k \,/\, (K \sigma_f^3)$	Third spectral moment of frequency deviation from centroid, weighted by magnitude.
Frequency kurtosis	$\sum_k (f_k - f_c)^4 S_k \,/\, (K \sigma_f^4)$	Fourth spectral moment of frequency deviation from centroid, weighted by magnitude.
Norm. bandwidth	$\sum_k \lvert f_k - f_c \rvert^{1/2} S_k \,/\, (K \sigma_f^{1/2})$	Half-order moment of spectral spread, normalised by $\sigma_f^{1/2}$.

Table 4: Top 10 features by combined Spearman–Pearson rank score with κ, for the AE channel (all sub-bands and broadband combined) and the UL channel. After the Stage 2 redundancy reduction, 32 AE and 6 UL features are retained for modelling; the retained sets are dominated by sub-band features for AE and amplitude/spectral features for UL.

AE (best feature per rank)			UL	
Feature	$\lvert\rho\rvert$	$\lvert r\rvert$	Feature	$\lvert\rho\rvert$
1–2 MHz complexity	0.891	0.736	Spectral mean	0.663
500k–1 MHz mobility	0.841	0.663	Peak	0.629
500k–1 MHz centre freq.	0.789	0.687	RMS	0.607
500k–1 MHz freq.-wt. std	0.789	0.686	Std	0.606
500k–1 MHz norm. freq. std	0.789	0.679	Spectral std	0.607
1–2 MHz shape factor	0.825	0.628	Variance	0.606
1–2 MHz spectral mean	0.801	0.590	Norm. bandwidth	0.607
1–2 MHz norm. bandwidth	0.782	0.599	Freq. skewness	0.539
1–2 MHz std	0.777	0.552	Freq. kurtosis	0.537
1–2 MHz RMS	0.777	0.552	Kurtosis	0.469

The top AE features are drawn overwhelmingly from the 1–2 MHz and 500 kHz–1 MHz sub-bands, with sub-band complexity ($|\rho| = 0.891$) and mobility ($|\rho| = 0.841$) heading the ranking. Broadband AE features rank lower than their sub-band counterparts, indicating that the frequency-specific content of the high-frequency AE signal carries more monotonic information about κ than the full-band aggregate. For the UL channel, amplitude features (spectral mean, peak, RMS, std) dominate at $|\rho| \approx 0.60$–0.66; the maximum UL correlation ($|\rho| = 0.663$) is substantially lower than the top AE correlation ($|\rho| = 0.891$). Figure 1 and Figure 2 show the full feature correlation rankings for each sensor.


AE — feature correlation ranking
AE_1000-2000kHz__complexity
AE_500-1000kHz__mobility
AE_500-1000kHz__center_frequency
AE_500-1000kHz__frequency_weighted_std
AE_500-1000kHz__normalized_frequency_std
AE_1000-2000kHz__shape_factor
AE_1000-2000kHz__spectral_mean
AE_1000-2000kHz__normalized_bandwidth
AE_1000-2000kHz__std
AE_1000-2000kHz__rms
AE_1000-2000kHz__spectral_std
shape_factor
AE_500-1000kHz__rms_frequency
AE_500-1000kHz__dominant_frequency
AE_500-1000kHz__peak
AE_1000-2000kHz__variance
spectral_skewness
frequency_skewness
AE_500-1000kHz__complexity
AE_1000-2000kHz__spectral_skewness
spectral_kurtosis
AE_500-1000kHz__variance
AE_20-500kHz__shape_factor
AE_1000-2000kHz__spectral_kurtosis
spectral_flatness
center_frequency
mobility
AE_500-1000kHz__spectral_std
AE_500-1000kHz__rms
AE_500-1000kHz__std
frequency_weighted_std
rms_frequency
AE_20-500kHz__center_frequency
AE_20-500kHz__frequency_weighted_std
AE_500-1000kHz__crest_factor
AE_20-500kHz__mobility
margin_factor
AE_500-1000kHz__impulse_factor
AE_20-500kHz__normalized_frequency_std

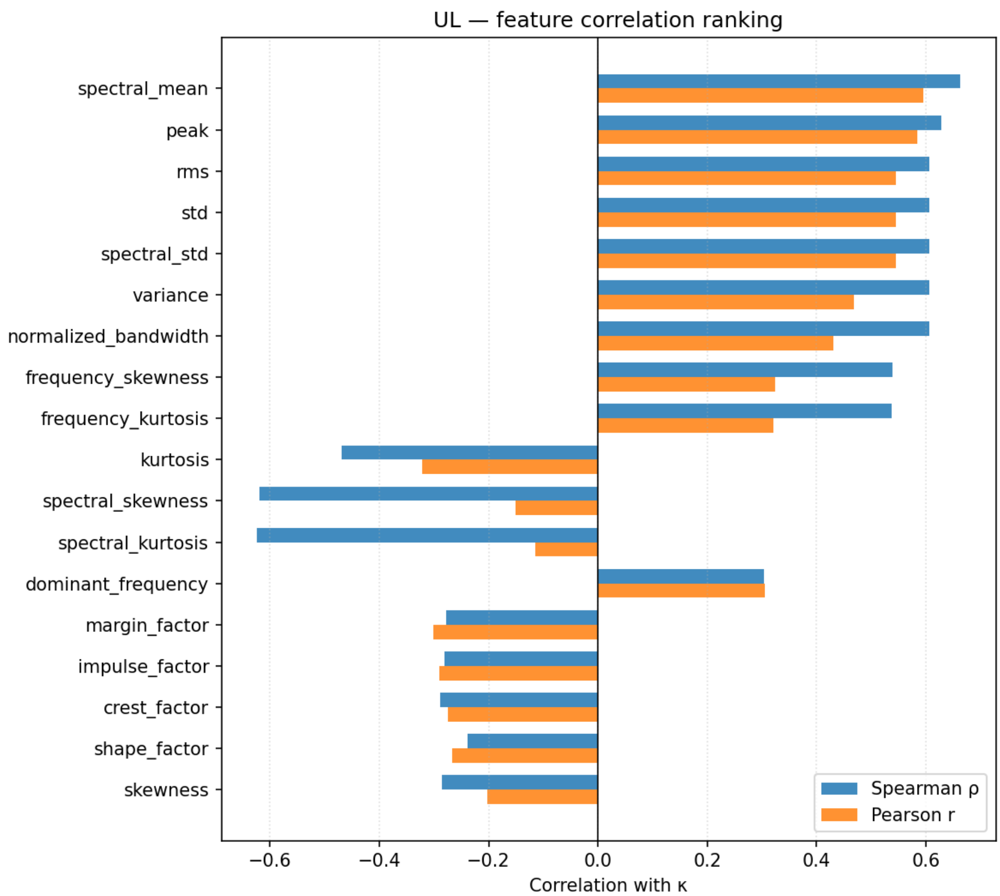


Figure 2: Spearman $|\rho|$ and Pearson $|r|$ between each UL-channel feature and κ, sorted by combined rank. Retained features highlighted.

5.2 Operating-condition confounding

The raw Spearman correlation between broadband signal energy (as a proxy for all amplitude features) and κ is substantially driven by the shared dependence on rotational speed and temperature. Quantitatively, RPM has a Spearman correlation of $\rho = +0.62$ with κ and temperature has $\rho = -0.68$. Higher RPM reduces the required viscosity ν_1 (Eq. (2)) and therefore increases κ, while simultaneously increasing signal energy through more frequent contact events per second. Higher temperature reduces the actual viscosity ν_0 and therefore reduces κ, while also affecting material damping and structural dynamics.

As a consequence, amplitude-dominated features such as RMS, variance, and peak — which rank among the top features by raw Spearman correlation — are partially acting as indirect proxies for rotational speed rather than as direct indicators of lubrication film adequacy. This confounding is an inherent property of any single-operating-point sensor system without an explicit operating condition normalisation step.

Spectral shape features (frequency kurtosis, normalised bandwidth, spectral flatness) are less directly tied to total signal power and are therefore expected to carry a higher proportion of genuine lubrication-regime information relative to their raw $|\rho|$ rank. The κ regression framework partially addresses this by training models on the full multi-condition dataset, where speed and temperature appear as independent axes of variation and the model can implicitly learn to discriminate amplitude changes due to lubrication from those due to operating conditions.

5.3 PCA and regime separability

Figure 3 shows the first two principal components of the retained AE feature set, coloured by κ regime. Boundary and full-film sweeps are broadly separated in the PC1–PC2 plane, while mixed regime sweeps overlap with both. This confirms that the retained feature set carries regime-relevant discriminative information but that the mixed lubrication regime remains the most challenging region, consistent with the known partial physical overlap between boundary and mixed conditions.

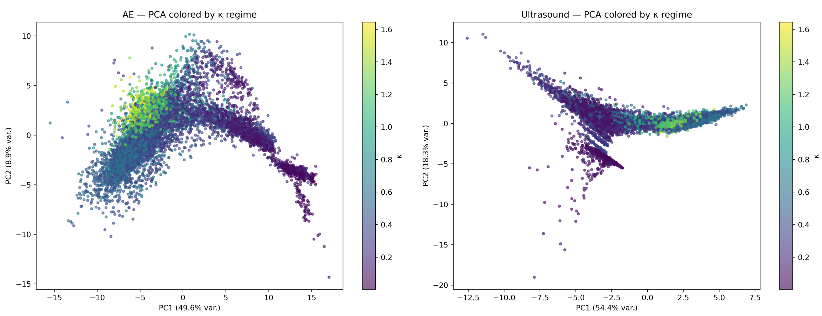


Figure 3: PCA projection of the retained AE feature set, coloured by lubrication regime. Boundary ($\kappa < 0.5$): red; mixed ($0.5 \leq \kappa < 1$): orange; full film ($\kappa \geq 1$): green.

6 Regression Modelling and Statistical Evaluation

6.1 Experimental design and holdout split

The dataset is divided into a training set (80%) and a hold-out test set (20%) by sweep-level random sampling, stratified to preserve the κ regime distribution. The hold-out set is withheld entirely from all model development steps and used only for final evaluation.

Three regression model families are evaluated for three sensor configurations (AE only, UL channel only, and combined AE+UL features):

1. **Elastic Net** (regularised linear regression with combined ℓ_1 and ℓ_2 penalty): hyper-parameters α and ℓ_1 ratio selected by `ElasticNetCV` with an inner cross-validation loop (9 α values, 6 ℓ_1 ratios).

2. **Degree-2 Polynomial Regression** (polynomial feature expansion followed by Ridge regression via `RidgeCV`): captures pairwise interactions between features while

remaining linear in parameters.

3. **LightGBM** (gradient-boosted decision trees): hyperparameters (learning rate, number of leaves, tree depth, minimum child samples, ℓ_2 regularisation) tuned by Bayesian optimisation using the Optuna TPE sampler (20 trials per inner fold).

6.2 Repeated nested cross-validation

To obtain stable performance estimates that properly account for the variance introduced by the train/validation split, model evaluation uses a repeated nested k-fold cross-validation scheme. The outer loop is repeated $R = 5$ times with independently re-seeded five-fold splits, yielding $R \times k = 5 \times 5 = 25$ outer RMSE scores per model–sensor combination. For LightGBM, each outer fold contains its own inner Optuna optimisation loop, ensuring that the reported outer scores are not contaminated by hyperparameter selection on the outer fold data. For the linear models, hyperparameter selection is handled by `ElasticNetCV` and `RidgeCV` within each outer training partition.

The 25 outer scores form the empirical sampling distribution of RMSE for each model and are used directly in the significance tests described below.

6.3 Statistical significance testing

Pairwise significance tests are conducted at two levels.

Level 1 — Within-sensor comparisons (architecture differences). For pairs of models trained on the same sensor configuration (e.g., Elastic Net vs. Polynomial on AE), the corrected repeated k-fold t-test of Nadeau and Bengio [12] is applied to the 25 paired RMSE scores. This test corrects the standard paired t-test for the positive correlation induced by shared training data across repeated folds. The correction factor is:

$$\hat{\sigma}^2_{\text{corr}} = \left(\frac{1}{n_{\text{test}}} + \frac{n_{\text{train}}}{n_{\text{test}}} \right) \hat{\sigma}^2, \tag{3}$$

where $\hat{\sigma}^2$ is the sample variance of pairwise RMSE differences and $n_{\text{test}}/n_{\text{train}}$ is the test/train size ratio. Multiple within-sensor comparisons are adjusted by the Holm–Bonferroni step-down procedure at $\alpha = 0.05$.

Level 2 — Cross-sensor comparisons (best per sensor configuration). For the best model from each sensor configuration (selected by lowest mean CV RMSE), pairwise differences on the common hold-out set are assessed by:

- The Wilcoxon signed-rank test on paired residuals (non-parametric);

- The modified Diebold–Mariano test [13] for equal predictive accuracy under squared-error loss;

- A bootstrap ΔRMSE 95% confidence interval (10 000 resamples with replacement).

Cross-sensor p-values are also corrected by Holm–Bonferroni.

6.4 Results

Table 5: Regression model performance. CV RMSE: mean ± std of outer RMSE across 25 repeated nested CV folds ($n_{\text{train}} = 7\,026$ per fold). HO: hold-out test set ($n = 1\,757$). Best hold-out R^2 in bold. Bayesian Ridge (BR) was trained on the full training set and evaluated on hold-out only (not included in the nested CV framework) and is shown for reference.

Model	Sensors	CV RMSE	HO R^2	HO MAE	HO RMSE
LightGBM	Combined	0.0767 ± 0.0040	**0.953**	0.050	0.076
LightGBM	AE	0.0780 ± 0.0037	0.951	0.051	0.078
LightGBM	UL	0.2233 ± 0.0060	0.600	0.155	0.222
Elastic Net	Combined	0.1556 ± 0.0051	0.830	0.111	0.145
Elastic Net	AE	0.1568 ± 0.0052	0.826	0.113	0.147
Polynomial	Combined	0.1568 ± 0.0048	0.803	0.113	0.156
Polynomial	AE	0.1646 ± 0.0044	0.775	0.123	0.167
Polynomial	UL	0.2577 ± 0.0056	0.477	0.193	0.254
Elastic Net	UL	0.2783 ± 0.0068	0.396	0.210	0.273
BR (ref.)	Combined	—	0.847	0.103	0.137
BR (ref.)	AE	—	0.830	0.110	0.145
BR (ref.)	UL	—	0.536	0.181	0.239

Overall performance. LightGBM achieves the best hold-out performance across all sensor configurations, with AE-only reaching $R^2 = 0.951$, RMSE $= 0.078$ κ-units. On a target range of [0.003, 1.64] with mean 0.51, this corresponds to a relative RMSE of approximately 15% of the target range. Linear and polynomial models on AE features achieve $R^2 \approx 0.77$–0.83, confirming a non-linear feature–κ mapping that gradient boosting exploits more effectively.

Sensor comparison. AE features substantially outperform UL-channel features across all model families. The LightGBM AE–UL gap ($\Delta R^2 = 0.35$) is confirmed as highly significant (Nadeau–Bengio $p < 10^{-22}$; Wilcoxon $p < 10^{-166}$; ΔRMSE $= 0.144$, 95% CI [0.133, 0.157]). This reflects the richer frequency content and more direct physical sensitivity of the AE sensor to asperity contact events.

Sensor fusion yields a marginal, statistically unconfirmed improvement. LightGBM on the combined AE+UL feature set achieves $R^2 = 0.953$, marginally above the AE-only result (0.951). The within-CV corrected t-test rejects equality ($p = 0.024$), but hold-out tests do not confirm the improvement: Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap ΔRMSE $= 0.0011$ with 95% CI $[-0.0009, 0.0033]$ spanning zero. For the linear models, the combined feature set is similarly on par with AE alone (Elastic Net:

R^2 0.830 vs. 0.826; Polynomial: 0.803 vs. 0.775). The marginal fusion gain is consistent with the UL channel contributing a small amount of non-redundant information at low-frequency.

Model consistency. The close agreement between nested CV RMSE and hold-out RMSE across all models indicates stable generalisation with no evidence of significant overfitting. The Bayesian Ridge reference results (hold-out only) align closely with Elastic Net, confirming that the linear model results are not sensitive to the choice of regularisation method.

Figure 4 shows predicted vs. actual κ for the hold-out set for all model families under the AE-only configuration.

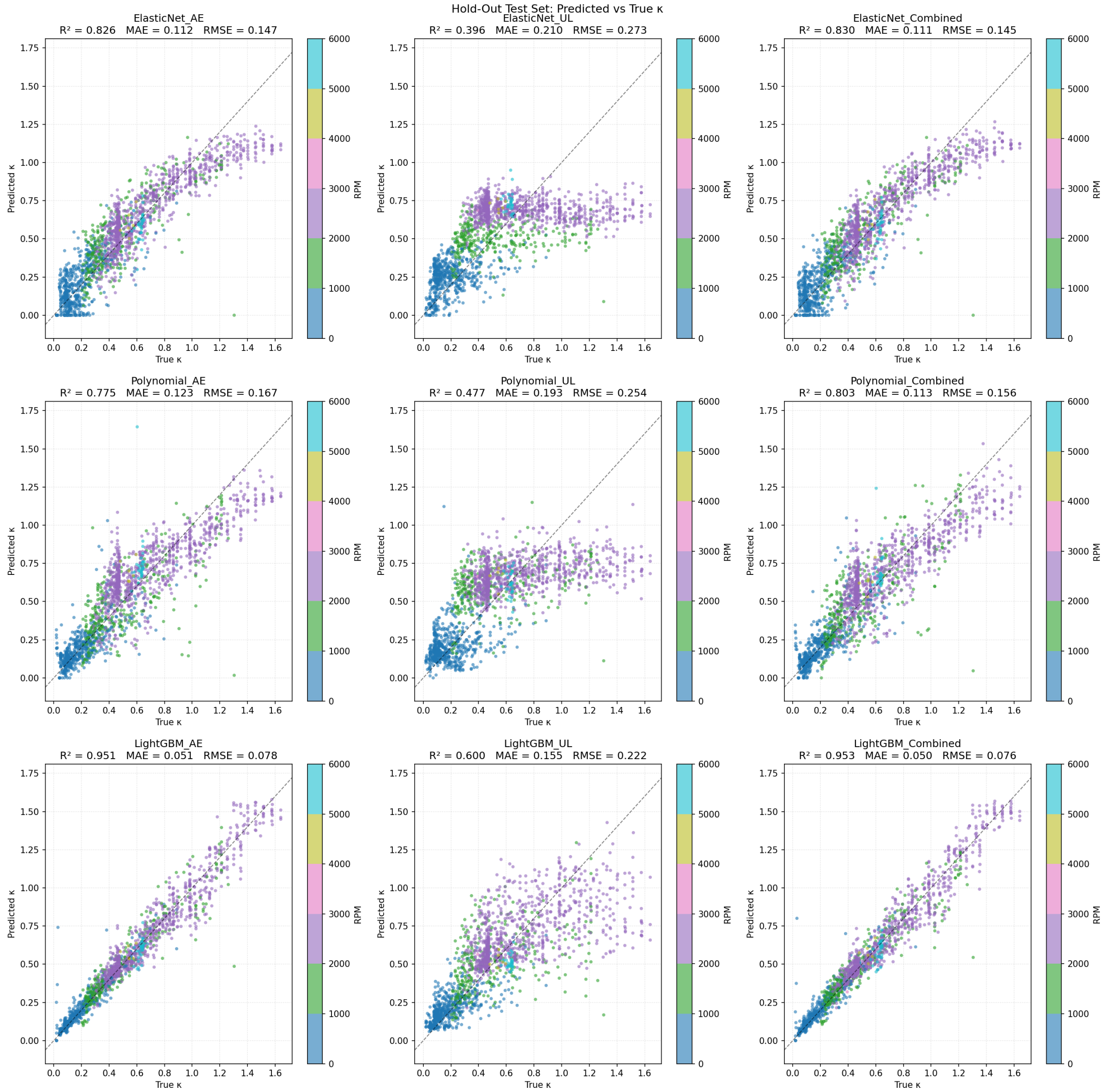


Figure 4: Predicted vs. actual κ on the hold-out test set for all model families and sensor configurations. The dashed diagonal indicates perfect prediction. Residual spread increases in the mixed lubrication regime ($0.5 \leq \kappa < 1.0$).

SHAP-based feature importance. SHAP (SHapley Additive exPlanations) values computed via TreeExplainer identify which AE sub-band features drive the κ predictions. The dominant contributor by a large margin is *1–2 MHz complexity* (mean |SHAP| = 0.162), followed by *500 kHz–1 MHz centre frequency* (0.051) and *1–2 MHz mobility* (0.038). The 1–2 MHz band complexity feature is also the only feature ranked in the top 10 across all three model types (cross-model agreement, Table 6), making it the most robust single predictor of κ in the dataset. Figure 5 shows the full importance ranking.

Table 6: Cross-model SHAP agreement for AE features: features appearing in the top 10 by mean |SHAP| for more than one model type.

Feature	ElasticNet	Polynomial	LightGBM	Models in top 10
1–2 MHz complexity	1	1	1	3
1–2 MHz mobility	2	–	3	2
500k–1 MHz spectral std	3	–	10	2
Broadband freq. skewness	10	–	4	2
500k–1 MHz centre freq.	–	10	2	2
500k–1 MHz dominant freq.	–	4	7	2

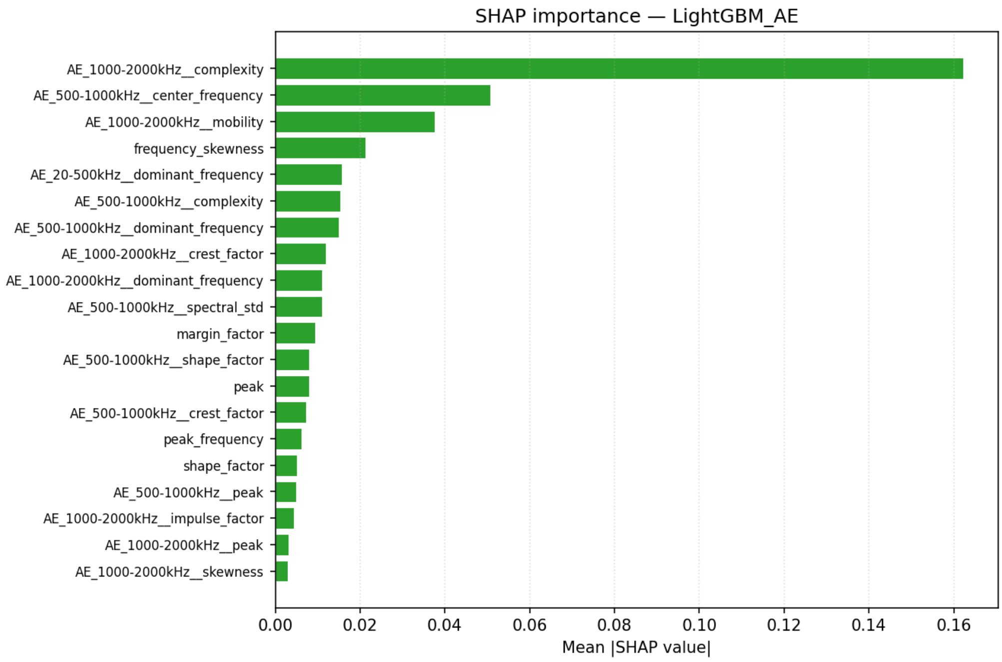


Figure 5: SHAP feature importance (mean |SHAP value|) for the LightGBM model trained on AE features. Features from frequency sub-bands are prefixed with the band label (e.g., `AE_20-500kHz_`).

7 Discussion

7.1 Why AE outperforms the UL channel

The consistently superior performance of AE over the UL channel across all model families reflects a fundamental hardware difference. The AE sensor captures genuine wideband stress-wave emissions up to 800 kHz, directly sensitive to individual asperity contact events. The UL channel is heterodyned, limiting its useful content to below approximately 10 kHz, where the signal is a mixture of low-frequency machinery noise, VFD harmonics, and structural vibration. The physical information about microscale contact events is largely absent in this frequency range. Replacing the heterodyned probe with a wideband passive acoustic or ultrasound sensor would be expected to substantially improve the UL-channel models.

7.2 Why fusion provides only marginal gain

The marginal fusion result ($\Delta R^2 = 0.002$ on the hold-out, not confirmed by Wilcoxon or Diebold–Mariano tests) can be understood from two complementary angles. First, the UL channel retains only 6 features after selection (vs. 32 for AE), and the top UL correlations ($|\rho| \leq 0.66$) are far weaker than the top AE correlations ($|\rho| = 0.89$). The UL channel therefore contributes a small amount of additional information that provides a marginal benefit to the combined model under CV, but the effect is too small to be detectable on the hold-out set (1 757 samples). Second, the UL channel's effective bandwidth ($<$ 10 kHz after heterodyning) means its information content partially overlaps with the low-frequency tail of the AE signal, limiting the independent contribution it can make.

This result is an informative contribution to the LCM literature, where AE–ultrasound complementarity is widely *argued* but rarely *tested*. Whether a wideband passive ultrasound sensor (rather than a heterodyned probe) would provide more independent information – and thus a more substantial fusion gain – remains an open question for future work.

7.3 Interpreting the amplitude-confounding result

The strong raw Spearman correlations of amplitude features (RMS, variance, Hjorth mobility) with κ should be interpreted with caution. As discussed in Section 5.2, rotational speed is the dominant driver of both κ (through ν_1 in Eq. (2)) and broadband signal energy (through the rate of contact events per revolution). A model trained on amplitude features therefore exploits speed as an indirect proxy for κ rather than sensing lubrication-specific signal changes directly.

This confounding does not invalidate the regression results: a model that uses speedmediated amplitude changes to predict κ is still predicting κ, and its predictions may be operationally useful. However, it does indicate that the model's generalisation to operating conditions outside its training distribution (different speed ranges, different bearing loads) may be limited. Future feature engineering should prioritise spectral shape and ratio features (frequency kurtosis, normalised bandwidth, spectral flatness) that are expected to carry a higher proportion of genuine lubrication-state information relative to operating condition proxies.

7.4 Implications for the κ regression framework

The κ regression framework is validated as a methodologically sound approach for passive non-invasive LCM. A hold-out $R^2 = 0.951$ and RMSE $= 0.078$ κ-units for a single AE sensor represents strong performance across a demanding continuous speed-and-temperature ramp spanning boundary through full-film lubrication conditions. This substantially exceeds the performance reported by Jakobsen et al. [1] in a comparable passive sensing study, which is likely attributable both to the richer sub-band feature set and to the much larger training set (7 026 vs. $\sim$640 sweeps) available from the continuous sweep protocol.

The repeated nested cross-validation design provides appropriately conservative performance estimates: the 25 outer RMSE scores capture fold-to-fold variance without the optimistic bias introduced by simple k-fold evaluation with shared hyperparameter search. The corrected significance tests confirm that performance differences between model families are assessed under assumptions that properly account for data re-use, addressing a common methodological weakness in the machine learning for condition monitoring literature.

7.5 Limitations

1. **Single bearing and lubricant.** All results are specific to the SYJ25 bearing and Keratech 22 oil. Generalisability to other bearing types, sizes, or lubricants is unknown and is a priority for follow-on experiments.

2. **Heterodyned UL channel.** The passive UL channel is frequency-limited to $\approx$10 kHz. It should not be equated with wideband ultrasound and results should be interpreted in light of this hardware constraint.

3. **Data gap in the mixed lubrication regime.** With only discrete temperature setpoints, the mixed regime ($\kappa \approx 0.5$–0.7) is underrepresented (24% of sweeps). This is precisely the regime of greatest practical interest for early lubrication deterioration warning.

4. **VFD harmonic contamination.** Switching artefacts at $\approx$5 kHz affect low-frequency spectral features of both channels. Features computed from the UL channel and from the AE band below 10 kHz should be interpreted accordingly.

5. **Model-derived κ reference.** No independent film thickness measurement (e.g., electrical capacitance or active ultrasonic reflectometry) is available. The κ values are model-derived and subject to the Hamrock–Dowson and Walther equation assumptions.

6. **Single session file.** Although 8 783 sweeps per sensor are retained, all data come from a single continuous speed-ramp session of one bearing. Temporal autocorrelation between adjacent sweeps may reduce the effective independence of the training and hold-out sets beyond what the sweep-level split guarantees. Validation across multiple independent sessions and bearings is needed.

8 Conclusion

This paper has presented an exploratory study of simultaneous acoustic emission and passive ultrasound sensing regressed to the ISO 281 viscosity ratio κ for non-invasive lubrication condition monitoring of a ball bearing. The main findings are as follows:

1. **AE sub-band features are strong predictors of κ.** LightGBM on 32 retained AE features achieves hold-out $R^2 = 0.951$, RMSE = 0.078 κ-units. The 1–2 MHz complexity feature is the dominant SHAP contributor and the only feature consistent across all three model types, identifying the high-frequency AE sub-band as the most informative frequency range for lubrication regime prediction.

2. **Sensor fusion provides only a marginal, statistically unconfirmed improvement.** The LightGBM combined model achieves $R^2 = 0.953$ ($\Delta R^2 = +0.002$ over AE alone), but hold-out significance tests (Wilcoxon $p = 0.88$, Diebold–Mariano $p = 0.28$, bootstrap 95% CI including zero) cannot confirm the improvement. The heterodyned UL channel contributes only 6 retained features with substantially weaker correlations than AE, limiting its independent contribution.

3. **AE strongly and significantly outperforms the UL channel.** The AE–UL gap ($\Delta R^2 = 0.35$ for LightGBM) is confirmed highly significant by all test statistics, reflecting the fundamental bandwidth limitation of the heterodyned UL probe.

4. **Rigorous evaluation via repeated nested cross-validation.** Twenty-five outer RMSE scores per model provide stable, conservative performance estimates; corrected significance tests confirm that LightGBM significantly outperforms linear

models within each sensor configuration, while the Elastic Net–Polynomial difference is not significant for the combined feature set.

5. **The κ regression framework is validated** as a highly effective approach for passive non-invasive LCM, achieving substantially higher performance than prior comparable studies, attributed to the sub-band feature set and the large continuous-sweep dataset.

Future work should address the identified limitations by: acquiring wideband passive ultrasound to replace the heterodyned probe; validating across multiple independent sessions, bearing types, and lubricants; investigating temporal autocorrelation and its effect on hold-out validity; and developing physically motivated sub-band ratio features that are explicitly designed to be robust to operating-condition variation.

CRediT authorship contribution statement

Frederik Appel Vardinghus-Nielsen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interests

The author declares that there are no competing interests.

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Data availability

The complete dataset (HDF5 signal files, extracted features in Parquet format, and the reproducible Python analysis pipeline) will be deposited on Zenodo upon acceptance [DOI to be assigned].

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted writing tools for language editing and structural drafting. After using these tools, the author reviewed and edited the content as necessary and takes full responsibility for the content of the publication.

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